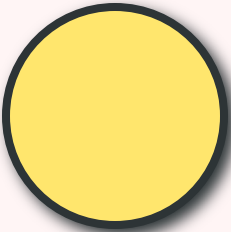
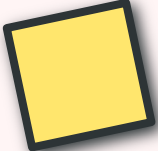


RESEARCH PRESENTATION

Reverse Thinking Makes LLMs Stronger Reasoners



REVTHINK: A Framework for Bidirectional Reasoning

 UNC Chapel Hill  Google Cloud AI  Google DeepMind

What We'll Cover Today

01

Introduction & Motivation

Understanding the importance of reverse thinking in AI reasoning systems

02

Related Work

Contextualizing REVTHINK within the broader research landscape

03

The REVTHINK Framework

A deep dive into methodology and architecture

04

Experimental Setup

Models, datasets, and evaluation methodology

05

Results & Analysis

Comprehensive evaluation and insights

06

Conclusion & Future

Key takeaways and research outlook

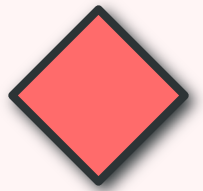
CHAPTER 01

Introduction & Motivation

Understanding the importance of reverse thinking
in AI reasoning systems

CORE CONCEPT

The Power of Reverse Thinking



→ Forward Reasoning

Process: Start with the problem and work step-by-step toward the solution

Example:

Emma has 2 apples, Jack has 3 apples.
 $2 + 3 = 5$ apples total

← Backward Reasoning

Process: Start from the solution and work backward to verify the problem

Example:

They have 5 apples total, Emma has 2.
Jack must have 3 apples ✓

💡 Why Bidirectional?

- 1 Consistency Checks:** Verify that forward and backward reasoning align
- 2 Error Detection:** Identify discrepancies that signal potential mistakes
- 3 Deeper Understanding:** Grasp inverse relationships between problems and solutions

Error Detection Example

✗ Incorrect Forward:

$2 + 3 = 6$ (Wrong!)

✓ Backward Check:

If total is 6, Emma has 2 → Jack has 4

Contradiction detected! Reassess solution.

Research Questions & Challenges

Q1 Beyond Math Domains

Can reverse thinking be applied to broader, less structured domains?

- ✗ **Prior Work:** Primarily focused on math reasoning
- ✗ **Math Advantage:** Highly structured with clear inverse relationships
- ✗ **Limitation:** Easy to create new problems by replacing variables

→ **Challenge:** Extend to commonsense, logic, NLI tasks

Q2 Training vs Test-Time

Can we train models to inherently think backward?

- ✗ **Current Approach:** Backward reasoning only at test time
- ✗ **Purpose:** Verification - check if forward reasoning is correct
- ✗ **Limitation:** Moderate improvements over self-consistency

→ **Challenge:** Internalize backward reasoning during training

⚠ Key Insight: The Gap in Current Methods

Domain Limitation

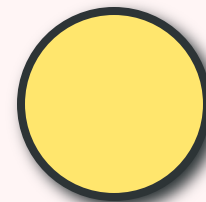
Mostly math-focused approaches

Test-Time Only

No training-time backward learning

Moderate Gains

Limited improvement over baselines



The REVTHINK Framework

Core Innovation

REVTHINK consists of data augmentation and novel learning objectives designed to instill reverse thinking in language models. **The key insight: internalize backward reasoning during training while maintaining zero-shot efficiency at test time.**

Stage 1: Data Augmentation

- 1** Generate Forward Reasoning
From teacher model with CoT
- 2** Generate Backward Question
Inverts the original problem
- 3** Generate Backward Reasoning
Solves the reversed question

Stage 2: Training Objectives

- A** $Q \rightarrow R_f$
Forward reasoning (standard)
- B** $Q \rightarrow Q_b$
Backward question generation
- C** $Q_b \rightarrow R_b$
Backward reasoning

⚡ **Test-Time Efficiency:** Only forward reasoning ($Q \rightarrow R_f$) is used at inference, making it as efficient as zero-shot approaches!

CHAPTER 02

Related Work

Contextualizing REVTHINK within the
broader research landscape



CURRENT APPROACHES

Reasoning with LLMs: The Landscape



Chain-of-Thought

Approach: Prompt LLMs to generate step-by-step reasoning before answering

Key Papers: Wei et al. (2022), Kojima et al. (2022)

Limitation: Single reasoning path, no verification



Self-Consistency

Approach: Sample multiple reasoning paths and select most consistent answer

Key Papers: Wang et al. (2022)

Limitation: Increased compute, no backward reasoning



Tree-of-Thought

Approach: Explore multiple reasoning paths systematically like a search tree

Key Papers: Yao et al. (2024)

Limitation: Expensive at test time, complex implementation



Self-Reflection

Approach: Models critique and refine their own outputs iteratively

Key Papers: Shinn et al. (2024), Madaan et al. (2024)

Limitation: Requires multiple inference passes



Multi-Agent

Approach: Multiple LLM agents collaborate and debate to reach consensus

Key Papers: Du et al. (2023), Liang et al. (2023)

Limitation: High computational cost, coordination challenges



Backward Reasoning

Approach: Verify solutions by reasoning backward from answer to question

Key Papers: Weng et al. (2022), Jiang et al. (2024)

Gap: Test-time only, mostly math, moderate gains

Knowledge Distillation: Core Concepts

What is Knowledge Distillation?

Knowledge distillation (KD) transfers knowledge from a **larger teacher model** to a **smaller student model**. The goal is to compress model size while preserving performance, making deployment more efficient.

Classic KD Approach

Teacher Distribution

$p_T(y|x)$ - soft probabilities



Minimize KL Divergence

$KL(p_S || p_T) \rightarrow$ align student to teacher



Student Distribution

$p_S(y|x)$ - matches teacher

Key Papers: Hinton et al. (2015), Buciluă et al. (2006)

Modern LLM KD Approaches

- 1 Direct Sampling**
Sample CoT rationales directly from teacher
- 2 Bootstrapping**
Generate reasoning via self-improvement
- 3 Multi-Teacher**
Combine outputs from multiple teachers

Key Insight for REVTHINK

Teacher models provide **Chain-of-Thought rationales** that can be sampled directly. REVTHINK extends this by also generating backward questions and reasoning, creating a richer training signal.

Dual Learning & Bidirectional Methods

↔ Dual Learning Concept

Core Idea: Leverage the primal-dual structure inherent to a task. The duality acts as **regularization during training**, enhancing performance across both tasks.

Machine Translation

English → German and German → English

Sennrich et al. (2016), Xia et al. (2018)

Dialogue Generation

Response generation and backward prediction

Lv et al. (2023), Shen et al. (2018)

Question-Answering

QA and question generation as dual tasks

Tang et al. (2017)

REVTHINK's Innovation

While dual learning is closely related, the dual relationships in prior studies are **relatively straightforward** (e.g., source-target language pairs).

REVTHINK's Novel Contribution:

Focuses on the **mutually inverse relationship** between a question and its backward counterpart

Key Differences

- ✓ **Backward questions and reasoning** are often absent and must be generated by LLMs
- ✓ **Establishes connections** between forward questions with forward reasoning and backward questions with backward reasoning
- ✓ **Exploits consistency** of this connection within training objectives

💡 REVTHINK incorporates backward question generation and backward reasoning as forms of regularization to improve reasoning capabilities

CHAPTER 03

The REVTHINK Framework

A deep dive into the methodology
and architecture

Framework Overview: Two-Stage Pipeline

1 Data Augmentation

1 Original Dataset

$D = \{(Q(i), A(i))\}$



2 Teacher Generation

Q, R_f, Q_b, R_b



3 Verification & Filtering

Keep only consistent pairs



Augmented Dataset D_{aug}

2 Student Training

A $Q \rightarrow R_f$

Generate forward reasoning

B $Q \rightarrow Q_b$

Generate backward question

C $Q_b \rightarrow R_b$

Generate backward reasoning

Multi-Task Learning

⚡ At test time, only forward reasoning ($Q \rightarrow R_f$) is performed, making test-time compute as efficient as zero-shot prompting!

FORMAL DEFINITION

Problem Setup & Notation

Dataset Definition

$$D = \{(Q(i), A(i))\}_{i=1}^n$$

A dataset of **n samples**, where each sample comprises:

Q Question - The input problem

A Answer - The ground truth

Teacher Model

We assume **black-box access** to a teacher model **T**:

- Can obtain outputs from **T**
- Cannot access internal logits
- Used for data generation

Backward Question Definition

A backward question is one that **reverses the original question**.

Forward Question (Q):

John has 3 apples, Emma has 2; how many total?



Backward Question (Q_b):

John and Emma have 5 apples total. If Emma has 2, how many does John have?

Augmented Dataset

$$D_{\text{aug}} = \{(Q(i), R_f(i), Q_b(i), R_b(i))\}$$

Each instance now contains 4 components

STAGE 1 DETAILS

Data Augmentation: Three-Step Process

1 Generate Forward Reasoning

Process: Prompt teacher T to generate Chain-of-Thought reasoning

$$R_f = T(Q; I_{fr})$$

Filtering:

Keep only if $g(R_f) = A$
(correct answer extraction)

2 Generate Backward Question

Process: Condition on Q and A to create inverse question

$$Q_b = T(Q, A; I_{bq})$$

Key Property:

Answer must be present in original Q

3 Generate Backward Reasoning

Process: Answer the backward question with CoT

$$R_b = T(Q_b)$$

Output:

Complete 4-tuple ready for training

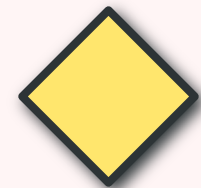
Prompt Design Examples

Multiple Choice Tasks:

"Generate an inverse question with {n} answer choices... Ensure only one answer choice is correct and present in the input question."

Math Reasoning Tasks:

"Generate an inverse question... Make sure there exists only one correct answer... The correct answer must be present in the input question."



Quality Control: Verification & Filtering

Two-Stage Filtering

1 Forward Reasoning Check

Verify $g(R_f) = A$ (correct answer extraction)

2 Backward Consistency Check

Verify backward reasoning aligns with original question

Consistency Verification

$$c = T(Q, A, Q_b, R_b; I_{con})$$

where $c \in \{0, 1\}$

c = 1

Consistent ✓

c = 0

Inconsistent ✗

Dataset Statistics After Filtering

StrategyQA 2,061 → 1,544

CommonsenseQA 9,741 → 6,478

ARC-challenge 1,199 → 1,035

MATH 7,500 → 2,511

GSM8K 7,379 → 4,293

ANLI (sampled) 100,459 → 883

i Despite significant filtering, verification improves data quality and downstream performance

Learning Objectives: Three-Task Training

Joint Loss Function

$$L = (1/3n) \sum [\ell(S(Q(i)), R_f(i)) + \ell(S(Q(i)), Q_b(i)) + \ell(S(Q_b(i)), R_b(i))]$$

where ℓ is cross-entropy between predicted and target tokens

A $Q \rightarrow R_f$

Generate Forward Reasoning

Student takes original question Q as input and generates forward reasoning R_f

Purpose:

Standard knowledge distillation - learn to solve problems

✔ Used at test time

B $Q \rightarrow Q_b$

Generate Backward Question

Student takes Q as input and learns to generate backward question Q_b

Purpose:

Learn to invert problems and determine right questions to ask

🎓 Training only

C $Q_b \rightarrow R_b$

Generate Backward Reasoning

Student takes Q_b as input and generates backward reasoning R_b

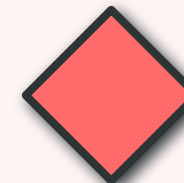
Purpose:

Reinforce ability to reason backward from solution

🎓 Training only

RATIONALE

Why These Objectives Work



A Forward Reasoning

Standard Knowledge Distillation

This is the **foundational objective** that teaches the student to solve problems step-by-step, following the teacher's reasoning patterns.

Why it works:

- Transfers teacher's reasoning patterns
- Teaches step-by-step problem solving
- Standard approach proven effective

B Backward Question

Learning to Invert Problems

This objective encourages the model to **"think" about how to invert a problem** and determine the right question to ask.

Why it works:

- Develops problem inversion skills
- Encourages deeper understanding
- Identifies key question components

C Backward Reasoning

Reinforcing Reverse Thinking

This objective **reinforces the student's ability to reason backward** from a solution to verify its correctness.

Why it works:

- Builds verification capabilities
- Strengthens bidirectional understanding
- Enables consistency checking

Multi-Task Learning Benefits

Auxiliary Tasks

Objectives B & C serve as regularization during training

Test-Time Efficiency

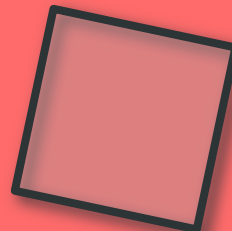
Only Objective A is used at inference

Internalized Skills

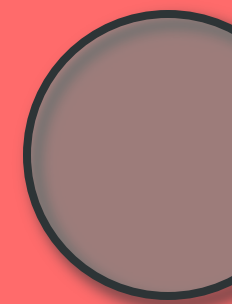
Backward thinking becomes inherent capability

CHAPTER 04

Experimental Setup



Models, datasets, and evaluation
methodology



CONFIGURATION

Models & Training Configuration

Teacher Model

Gemini-1.5-Pro-001

Google's advanced multimodal LLM

Student Models

Mistral-7B-Instruct-v0.3

Open-source instruction-tuned model

Gemma-7B-Instruct

Google's lightweight open model

Training Details



Fine-Tuning Method
LoRA with rank 32



Inference Engine
vLLM with greedy decoding



Temperature
0 (deterministic)

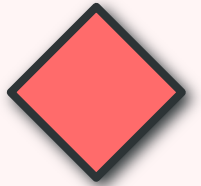


Epochs
3 for math, 10 for others

Learning Rates

Mistral-7B:
5e-6

Gemma-7B:
2e-4



Datasets: 12 Diverse Reasoning Tasks

Training Datasets (8)

Commonsense Reasoning

StrategyQA, CommonsenseQA, ARC-challenge

Mathematical Reasoning

MATH, GSM8K

Tabular Data Reasoning

TabMWP

Natural Language Inference

ANLI (r3)

Logical Reasoning

Date Understanding

Held-Out Datasets (4)

For generalization testing - no training data used

BoolQ

Yes/no questions (3,270 test)

OpenbookQA

Science questions (500 test)

e-SNLI

NLI with explanations (9,824 test)

GSM8K-Reversal

Reversed math problems (777 test)



Comprehensive evaluation across commonsense, math, logic, and NLI tasks ensures robust assessment of bidirectional reasoning capabilities

Baseline Comparisons

1 Zero-Shot

Approach: Direct prompting without any fine-tuning

Purpose:

Baseline reference for student model's raw capability

Method:

Kojima et al. (2022)

2 Knowledge Distillation

Symbolic KD (SKD)

Generate CoT from teacher, apply next-token prediction

Li et al. (2023), West et al. (2022)

Distill Step-by-Step

Add label prediction loss alongside CoT rationale

Hsieh et al. (2023)

3 Data Augmentation

Question Rephrasing

Paraphrase questions to create new ones
Yu et al. (2024)

Question Augmentation

Generate new questions based on original
Li et al. (2024)

Answer Augmentation

Sample more correct reasoning chains
Yu et al. (2024)

★ REVTHINK differs by introducing bidirectional thinking through backward question generation and backward reasoning objectives

CHAPTER 05

Results & Analysis

Comprehensive evaluation
and insights



KEY FINDINGS

Main Results: Superior Performance

🏆 Mistral-7B Results

StrategyQA	63.76% → 70.97%
------------	-----------------

CSQA	71.92% → 75.76%
------	-----------------

GSM8K	56.16% → 60.88%
-------	-----------------

Date Understanding	48.50% → 70.40%
--------------------	-----------------

+13.53% vs Zero-Shot
+6.84% vs Best Baseline

🏆 Gemma-7B Results

StrategyQA	56.77% → 64.19%
------------	-----------------

CSQA	73.01% → 74.53%
------	-----------------

GSM8K	52.24% → 57.21%
-------	-----------------

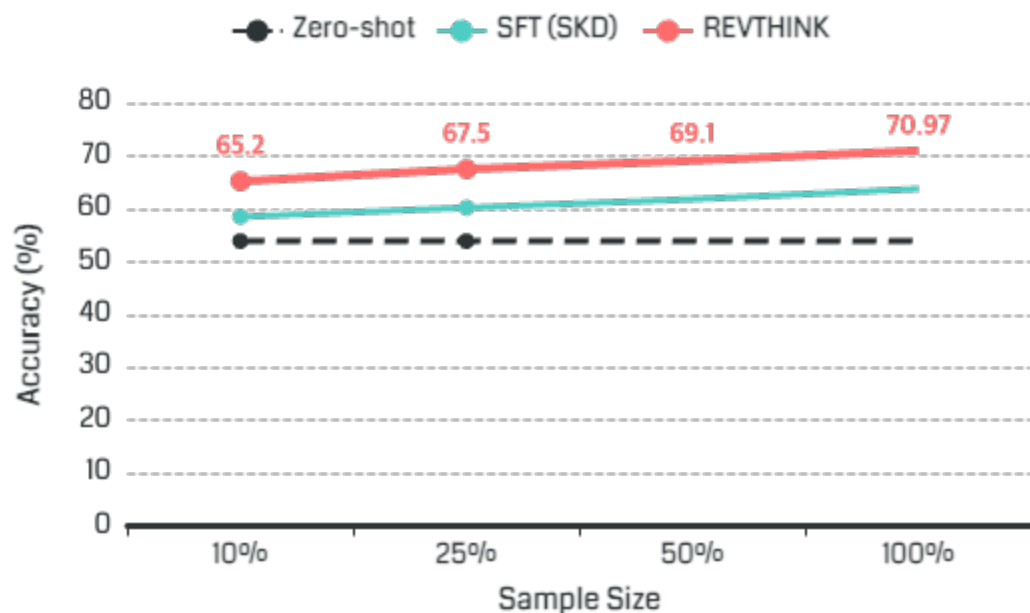
Date Understanding	59.62% → 66.27%
--------------------	-----------------

+14.37% vs Zero-Shot
+6.44% vs Best Baseline

★ REVTHINK consistently outperforms all baselines across diverse domains - commonsense, math, logic, and NLI tasks

Sample Efficiency: Less Data, Better Results

Performance vs Training Data Size



Key Findings

10% Data Achievement

REVTHINK with 10% training data outperforms SFT with 100% data on StrategyQA

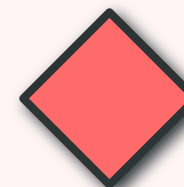
Consistent Advantage

REVTHINK outperforms SKD at **all data levels** (10%, 25%, 50%, 100%)

Scaling Behavior

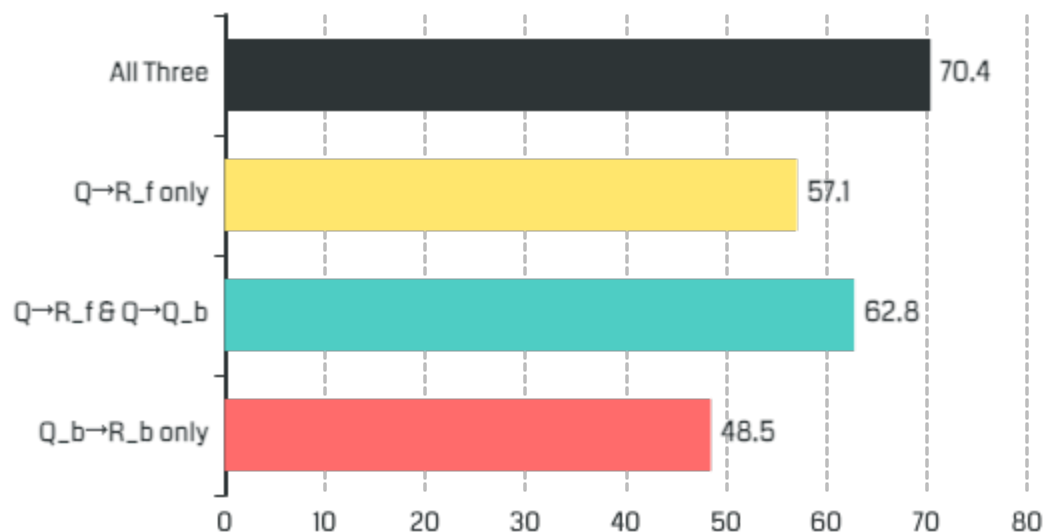
While SKD stagnates on some tasks, REVTHINK shows **clear upward trends**

⚡ Using just 10% of correct forward reasoning outperforms standard fine-tuning on 10x more data!



Component Analysis: What Makes REVTHINK Work

Learning Component Breakdown



Critical Findings

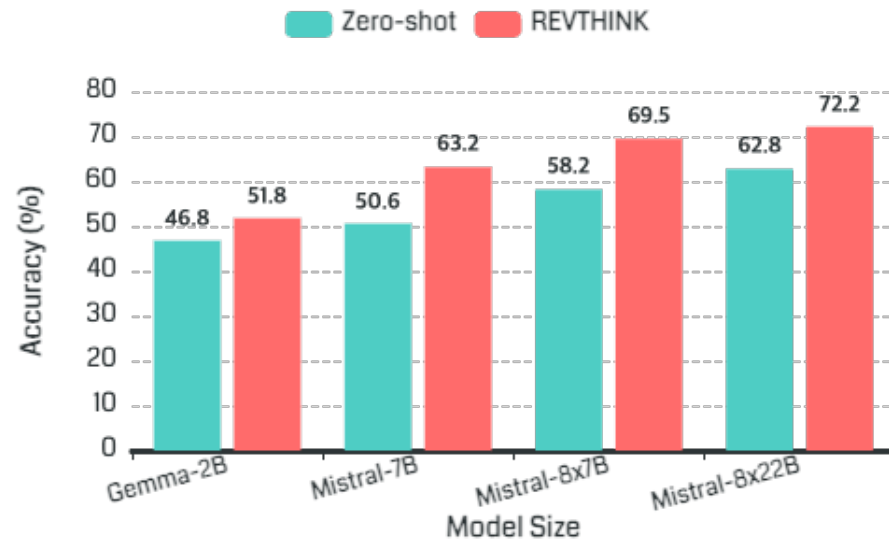
- ✗ Only $Q_b \rightarrow R_b$ Hurts Performance
Focusing solely on backward reasoning can fall below zero-shot
- ✓ $Q \rightarrow Q_b$ Already Helps
Adding backward question generation improves results
- ★ All Components Best
Full model with all three objectives performs optimally

Learning Variations (Date Understanding)

Multi-task (Instruction)	64.36%
Multi-task (Task-Prefix)	66.12%
Joint Objectives (REVTHINK)	70.40%

Model Scaling: Bigger is Better

Performance Across Model Sizes



Remarkable Achievement

Mistral-7B + REVTHINK

outperforms

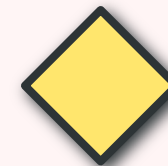
Mistral-8x22B Zero-Shot

by **8.36%**

Despite having 25× fewer parameters!

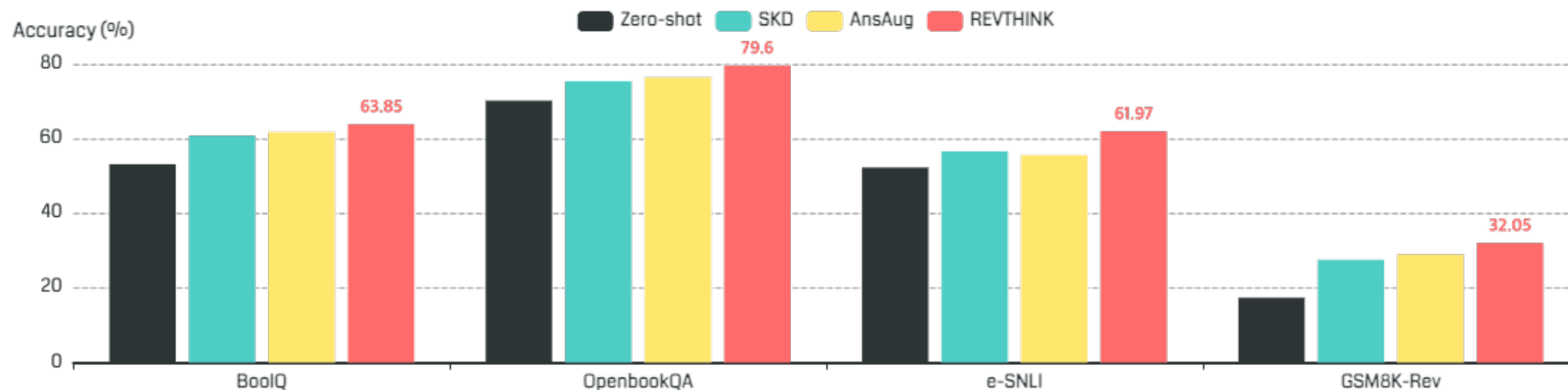
Consistent Improvements

Gemma-2B	+5.0%
Mistral-7B	+17.0%
Mistral-8x7B	+19.5%
Mistral-8x22B	+14.9%



Generalization: Out-of-Distribution Performance

Held-Out Dataset Results



Superior Generalizability

BoolQ: +2.11% vs AnsAug

OpenbookQA: +3.20% vs AnsAug

e-SNLI: +5.35% vs AnsAug

GSM8K-Rev: +3.09% vs AnsAug

Key Insight

Learning to reason backward enhances not just in-domain reasoning but also generalization to unseen datasets.

This indicates:

- Better problem understanding
- Reduced overfitting
- More robust reasoning capabilities

Complementary Strengths: Combining Methods

Performance Comparison

AnsAug Only

66.38%

StrategyQA - samples more forward reasoning

REVTHINK Only

70.97%

StrategyQA - enables reverse thinking

Combined (AnsAug + REVTHINK)

73.56%

StrategyQA - synergistic improvement!

Why They Complement

AnsAug Strength

Samples more correct forward reasoning chains, increasing training data diversity

REVTHINK Strength

Enables reverse thinking through backward question generation and reasoning

Combined Effect

More diverse forward data + bidirectional training = superior performance

GSM8K Results

AnsAug

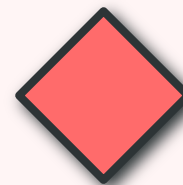
59.08%

REVTHINK

60.88%

Combined

61.58%



Training Efficiency: Token Count Analysis

Training Tokens vs Test Accuracy



Key Observations

Slightly More Training Tokens

REVTHINK produces **slightly more tokens** than SKD during training (due to backward components)

Comparable to AnsAug

Similar token count to Answer Augmentation but **much better performance**

Superior Efficiency

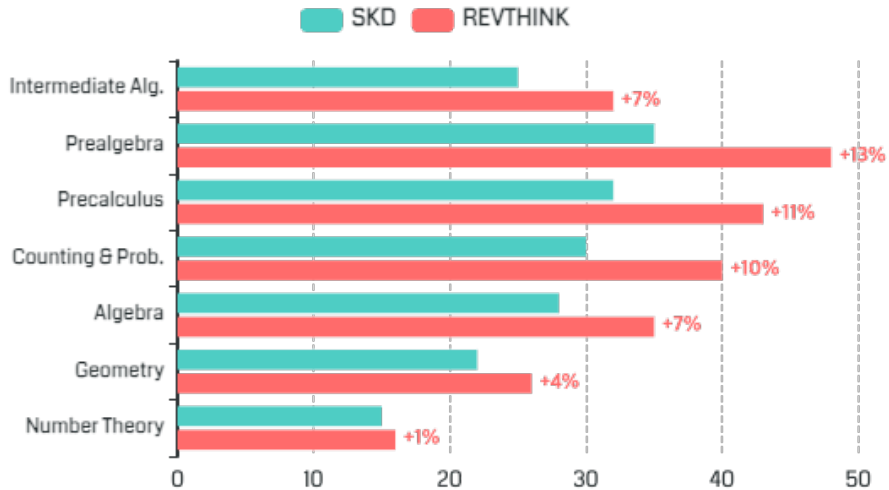
REVTHINK's deviation from regression line shows **better accuracy per token**

Test-Time Efficiency

At test time, REVTHINK generates **comparable number of tokens** to all baselines since only forward reasoning is used.

Problem Type Analysis: When REVTHINK Excels

MATH Dataset Breakdown



Greatest Improvements

Prealgebra

Clear inverse relationships between problem components

Precalculus

Well-defined functions with invertible operations

Counting & Probability

Problems with symmetric structures

Limited Improvement

Number Theory

Less invertible problems - thinking backward doesn't offer much advantage, or problems cannot be easily inverted

Difficulty Level Analysis

REVTHINK shows **greatest improvement on Level 3 (medium-hard) problems**, indicating that bidirectional reasoning is most beneficial for problems that are challenging enough to benefit from verification but not so difficult that backward reasoning becomes intractable.

CHAPTER 06

Conclusion & Future Directions

Key takeaways and
research outlook

SUMMARY

Key Contributions & Impact

1

First Bidirectional Framework

First framework to teach smaller models **backward reasoning across diverse tasks** beyond just math, including commonsense, logic, and NLI.

2

Effective Data Augmentation

Novel method for generating **structured forward-backward data** from teacher models with quality verification.

3

Multi-Task Learning Objectives

Three complementary objectives that **internalize reverse thinking** during training while maintaining test-time efficiency.

4

Test-Time Efficiency

Maintains **same efficiency as zero-shot** approaches at inference while delivering significant improvements.

5

Strong Empirical Results

13.53% average improvement over zero-shot, **sample efficiency**, positive scaling, and excellent generalization.

Performance Highlights

13.53%

Avg Improvement

10%

Data Efficiency

CONSIDERATIONS

Limitations & Considerations



Teacher Model Biases

Student models can **inherit biases from teacher models**. REVTHINK models share the same risks of misuse as other LLM-derived methods.



Task-Dependent Performance

Performance varies by problem type. Less effective on **non-invertible problems** like number theory where backward reasoning doesn't provide clear advantages.



Significant Data Filtering

Verification step **reduces training data significantly** :

ANLI: 100,459 → 883 samples
MATH: 7,500 → 2,511 samples



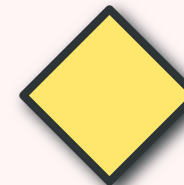
Prompt Engineering

Requires **careful prompt engineering** for backward question generation. Different task types need tailored prompts to ensure quality backward questions.

i Further research needed to effectively evaluate and mitigate biases in LLMs

LOOKING AHEAD

Future Research Directions



Extend to New Domains

Apply bidirectional reasoning beyond traditional reasoning tasks to:

- Creative writing and storytelling
- Code generation and debugging
- Multi-modal reasoning tasks
- Scientific hypothesis generation



Adaptive Objective Weighting

Investigate dynamic weighting of the three objectives based on:

- Task characteristics and complexity
- Problem invertibility measures
- Training progress and convergence
- Student model's current capabilities



Iterative Refinement

Explore iterative approaches where models:

- Alternate between forward and backward reasoning
- Self-correct based on consistency checks
- Progressively refine solutions
- Build verification into generation process



Combine with RLHF

Integrate with reinforcement learning:

- Use human feedback on reasoning quality
- Reward consistent bidirectional reasoning
- Penalize inconsistent forward-backward pairs
- Optimize for both correctness and coherence



Better Verification

Develop improved verification methods:

- Retain more training data
- Reduce false negatives in filtering
- Multi-step consistency checks
- Automated quality scoring

Exciting Opportunities

The bidirectional reasoning paradigm opens new avenues for improving LLM capabilities across diverse applications!



Thank You!

Questions & Discussion



Paper

arXiv:2411.19865



Code

Available on GitHub



Contact

cychen@cs.unc.edu

“Invert, always, invert.”

— Carl Jacobi